

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Tech.Branch :Computer Engineering/Computer Science & Engineering

Subject Code & Name: (BTCOE704C) Blockchain Technology Semester :VII

Max Marks: 60

Date: 08/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Draw and explain the architecture of a block in a blockchain.	Understand	6
B) Write a short note on following : i)Markle Tree ii)Digital Signature	Understand	6
C) Illustrate the basic operations carried out in public-key cryptography.	Understand	6
Q.2 Solve Any Two of the following.		12
A) Write a short notes on following : i)Hashcash PoW ii)Bitcoin PoW	Understand	6
B) Explain the benefits and limitations of bitcoin cryptocurrency.	Apply	6
C) Explain bitcoin mining with the help of a neat diagram.	Understand	6
Q.3 Solve Any Two of the following.		12
A) Explain the Lamport-Shostak-Pease BFT algorithm.	Understand	6
B) Define Consensus. Explain in detail the distributed consensus in a closed environment.	Understand	6
C) Write difference between permissioned and permission less blockchain.	Analysis	6
Q.4 Solve Any Two of the following.		12
A) Explain how it is achievable to mortgage using the blockchain.	Apply	6
B) Write the overview of supply chain finance.	Understand	6
C) What are the advantages of blockchain technology applied in the food industry?	Apply	6
Q.5 Solve Any Two of the following.		12
A) Define Smart contract. Explain the process of writing smart contracts. using Ethereum.	Understand	6
B) Define the workflow of Hyperledger fabric.	Understand	6
C) What is membership and access control in blockchain?	Understand	6

***** End *****